LECTURE-24

Monday 3/5/28.

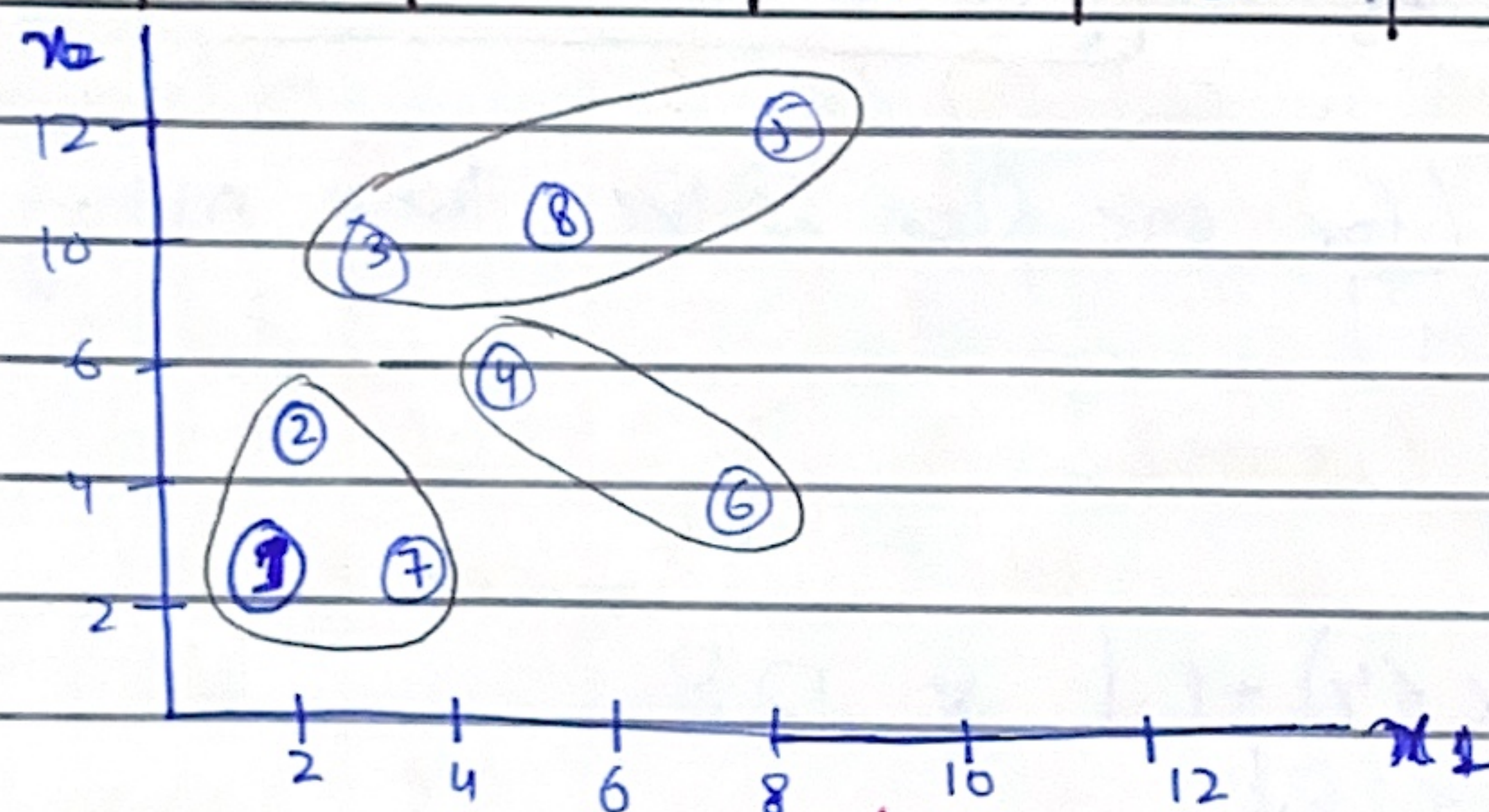
1 Clustering:

K=3 : 3 clusters.

Sr.	x_1	x_2	D(C1)	D(C2)	D(C3)	Itera. 1 Clusters	Itera. 2 Clusters
1 C1	1	3	0	6	11	C_1	C_1
2	2	5	3	3	8	C_1	C_1
3	3	10	9	7	2	C_3	C_3
4 C2	4	6	6	0	5	C_2	C_2
5	7	12	15	9	4	C_3	C_3
6	8	4	8	6	9	C_2	C_2
7	3	3	2	4	9	C_1	C_1
8 C3	5	10	11	5	0	C_3	C_3

Using Manhatten
$|1-5|+|3-10|$
$=11$.

★



- Make any random three Sr. (since K=3) as three cluster centers (C1, C2, C3).
- First we'll find the distance (eucledean or manhatten) b/w each row's (x_1, x_2) and the 3 clusters centers made (Sr. 1, Sr. 4, Sr. 8)
- Then we'll see with which cluster center it has the least distance; uss cluster main main vo orderded pair of input aa jaye ga and we'll write its designated cluster in "cluster" column.
- Commitle clusters for all rows
- Mark cluster circles on graph.
- Find cluster centers of each cluster.
- Do this by taking average of x and y values in each cluster.

C1	C2	C3
1, 3	4, 6	3, 10
2, 5	8, 4	7, 12
3, 3	12/2, 10/2	5, 10
6/3, 11/3	Clust. center = (6, 5)	15/3, 32/3
cluster center = (2, 3.6)		C.C = (5, 10.6)

- Now we've new cluster centers after taking averages.
- Now we'll run a second iteration & find distances from each ordered pair (row) to each new cluster center.
- Then with the new distances; find lowest distance from each cluster and allocate that cluster to that row.
- If we've same clusters made as first iteration; then we'll stop. Otherwise keep finding new cluster centers until no change in clusters.

Agglomerative Clustering:

$|1-2|+|3-5| = 3$

	P1	P2	P3	P4	P5	P6	P7	P8
P1	0							
P2	3	0			Symmetric Here!!			
P3	9	6	0					
P4	6	3	5	0				
P5	15	12	6	9	0			
P6	8	7	11	6	9	0		
~~P7~~	~~2~~	~~3~~	~~7~~	~~4~~	~~13~~	~~6~~	0	
P8	11	8	2	5	4	9	9	0

dist.	K	Clusters
–	8	P1, P2, P3, P4, P5, P6, P7, P8
2	7	(P1, P7), P2, P3, P4, P5, P6, P8
2	6	(P1, P7), (P3, P8), P2, P4, P5, P6
	5	
	⋮	
	1	

- Find each point's distance (manhatten or euclidean) with every other point
- Then find the least distance in the entire grid which is '2' at (P1, P7) and (P3, P8). Take any one cordinate to merge (we took P1, P7 here).

- Now we'll committe distance of (P_1, P_7) from each point and write all values in shrinked 2D new matrix which has row 7 deleted and P_7 merged with P_1.

- P_1 P_2 ....
$(P_1 P_7)$:
P_2 :

The distance between (P_1, P_7) and any point P could be committed by three ways:

during with max min. ← Max, min, Avg (will be given in quest)

- $P_1 \rightarrow P_3 = 9$
$P_7 \rightarrow P_3 = 7$

Now in the cell $(P_1, P_7) \leftrightarrow P_3$; we'll write '9' as it's max b/w two values.

- Refill the values of these in shrinked table's row of (P_1,P_7) and now find again the lowest value in entire table.
- In this case; the values of row (P_1, P_7) didn't change ∴ we still have lowest value = 2 of (P_3, P_8) so we'll merge these.
- Now shrink grid by one row and make them 6 rows. & committe distances of (P_3, P_8) from each point again.
- we'll keep finding min. table value & form clusters UNTIL we're k=1 in (prev 'DKC' table) means only one cluster.